

Transformer (IceCube 2D) Plots (1FA006)

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The Jupyter notebook can be found in this [GitHub repository](#).

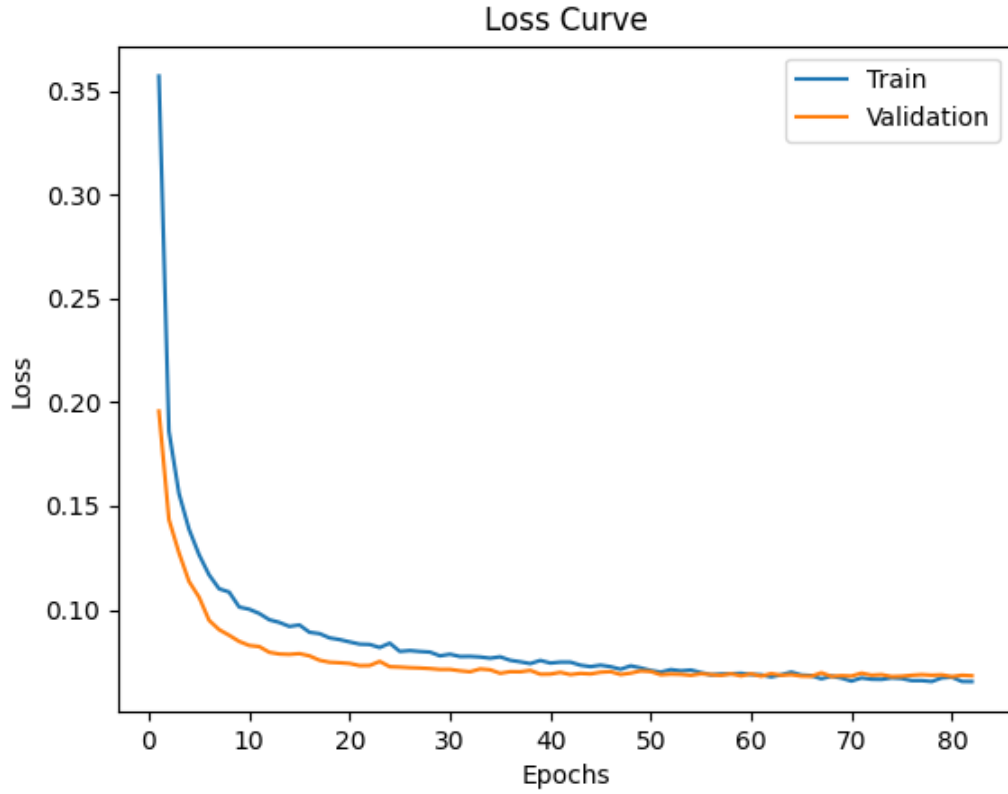


Figure 1: Loss curve of the Transformer model. This shows that training is stable and saturated at around epoch 50 and kept oscillating around the same value till epoch 82. The generalization error is almost zero near saturation. It was interesting to see the validation loss less than the training loss for a change.

Bivariate Density Plot

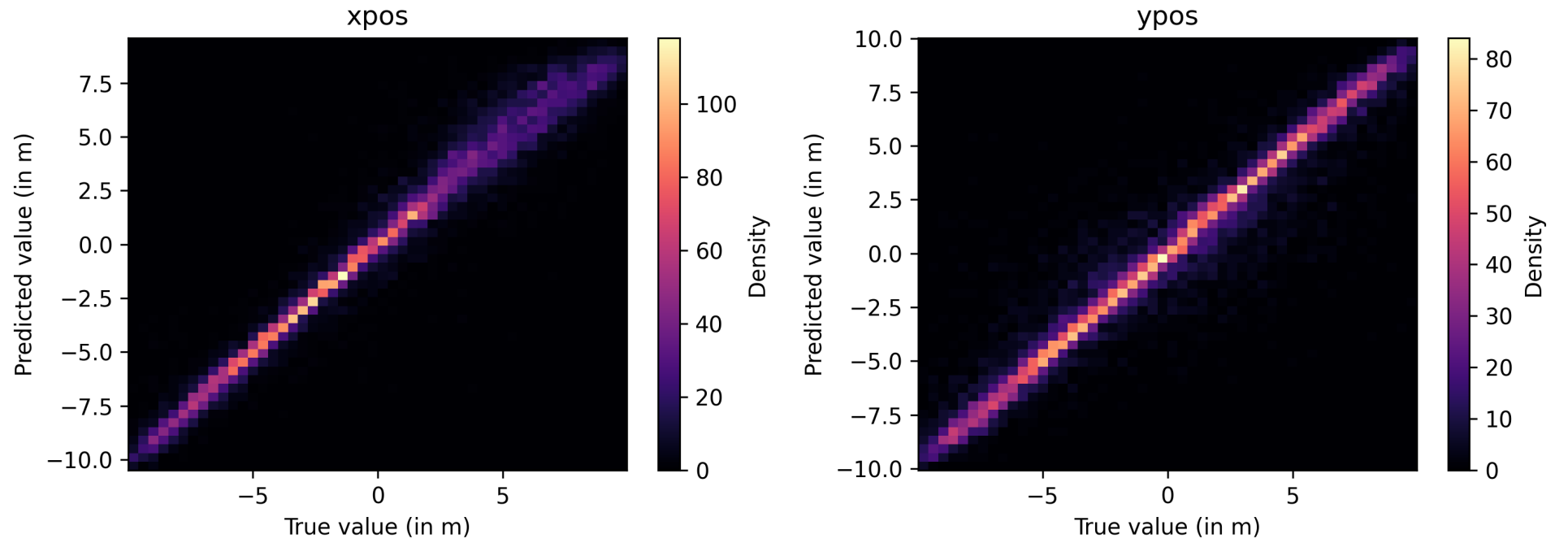


Figure 2: Predicted v/s True heatmap to show the model performance. Both the x_{pos} and y_{pos} show a clear $y = x$ trend with y_{pos} being more clearer and x_{pos} is more scattered at higher positive values. Comparing to the GNN plots, the scatter plot looks more sharp and narrow.

Residual Distribution

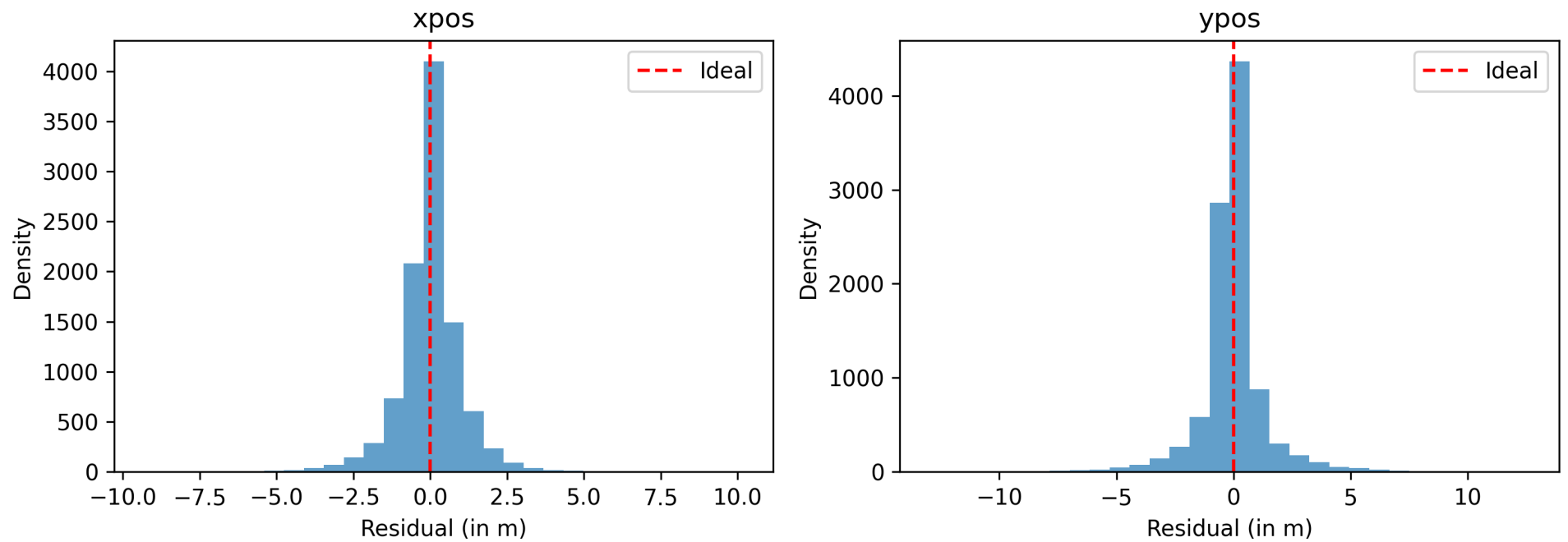


Figure 3: Residual Distributions. Both the distributions are centered around zero with a sharp peak. This means that most of the test samples have residuals (absolute value) less than 2.5m.

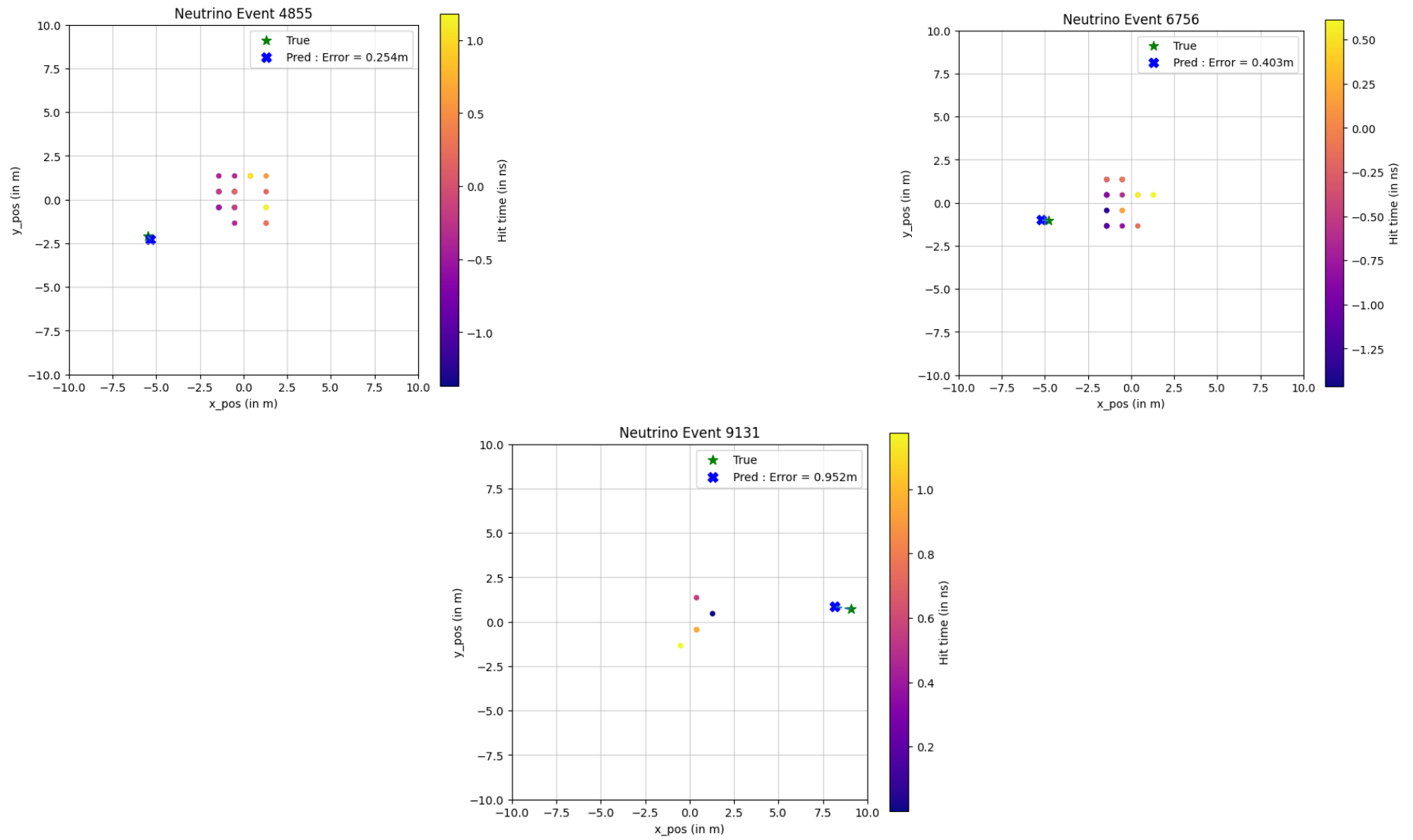


Figure 4: Examples of neutrino event positions along with the detector hit position and time, and predicted position.